

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Fuel System Clean Care Kit, Part Number 4919073.

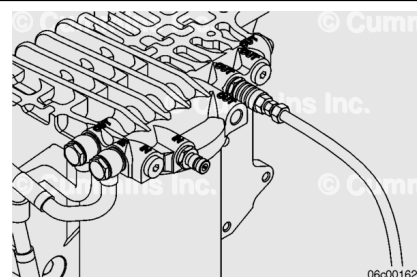
Additional Service Items

- No additional service items required.

General Information

The fuel filter head and engine control module (ECM) mounting plate are combined into a single casting.

Note : The ports on the fuel filter head are labeled to indicate which port is the inlet or outlet side of the fuel filter.



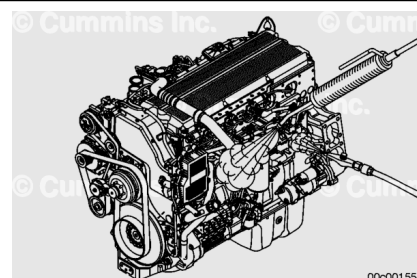
Preparatory Steps

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ CAUTION ⚠

Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.

Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.) which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

It is important to cover, cap or plug any openings as soon as possible when servicing the fuel system in order to prevent engine damage. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ CAUTION ⚠

Drain the fuel-water separator into a container, and dispose of contents in accordance with local environmental regulations.

- Disconnect the battery. See equipment manufacturer service information.
- Remove the pressure-side fuel filter. Refer to Procedure 006-065 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-065-tr.html)
- Remove the ECM. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/377/377-019-031.html)

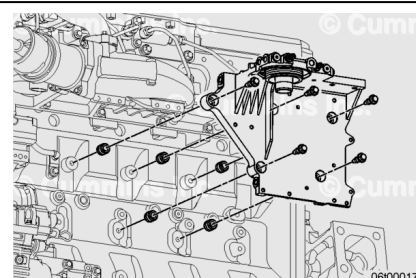
- Disconnect the ECM cooling plate and high-pressure fuel pump fuel supply lines. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-024.html>)
- Disconnect the fuel return line connections. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-013.html>)
- Remove the fuel drain manifold. Refer to Procedure 006-021 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-021.html>)
- If equipped, remove the ECM cooling plate mounted lift pump. Refer to Procedure 005-045 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-045.html>)

Remove

Note : The capscrew through the lift pump support must be removed.

Note : Vibration isolators are located behind the ECM mounting plate.

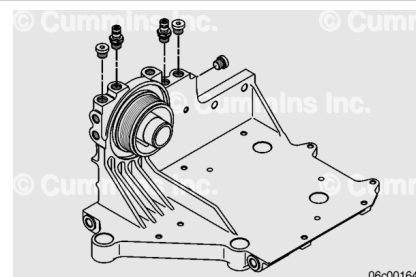
Remove the fuel filter head mounting capscrews.



Disassemble

Remove the Compuchek™ and threaded plugs from the fuel filter head.

Inspect the Compuchek™ fitting and threaded plug o-rings for damage.

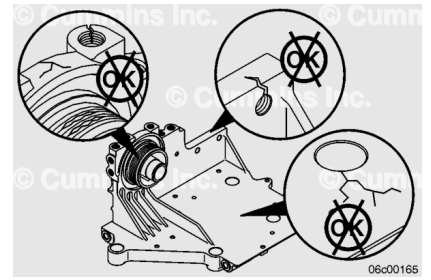


Inspect for Reuse

Inspect the fuel filter head and ECM cooling plate for cracks, passage blockage, and material or debris on the sealing surfaces.

Inspect the vibration isolators for damage.

Inspect the capscrews mounting the filter head to the block for damage to the sleeve or threads.

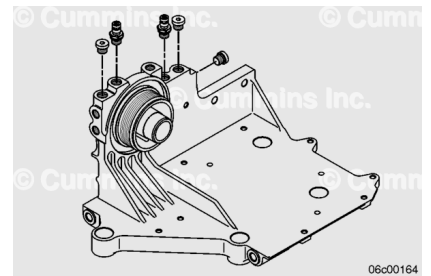


Assemble

Install the Compuchek™ and threaded plugs in the appropriate fuel filter head ports.

Torque Value: 20 n•m [177 in-lb]

Install the lift pump. Refer to Procedure 005-045 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-045.html>)



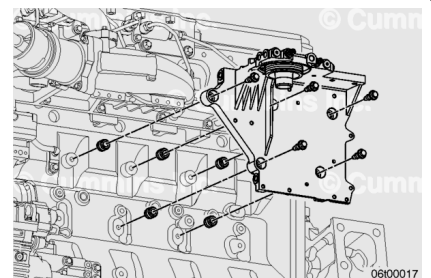
Install

Note : The capscrew through the lift pump support **must** be installed.

Install the fuel filter head capscrews through the filter head casting and vibration isolators.

Tighten the retaining capscrews.

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

**Hold the banjo fittings while tightening the banjo bolt to prevent the fitting from rotating.
Allowing the banjo fitting to rotate may damage the fuel line.**

- Install the fuel drain manifold. Refer to Procedure 006-021 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-021.html)
- Connect the fuel drain lines. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)
- Connect the ECM cooling plate and high-pressure fuel pump fuel supply lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/377/377-006-024.html)
- Install the ECM. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/377/377-019-031.html)
- Install the pressure-side fuel filter. Refer to Procedure 006-065 in Section 6.
(/qs3/pubsys2/xml/en/procedures/377/377-006-065-tr.html)
- Connect the battery. See equipment manufacturer service information.
- Operate the engine. Check for leaks.